

Furqaan Khurram Qamar

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EDUCATION

B.Eng. Mechatronics Engineering (Co-op) | McMaster University

Hamilton, Ontario

o **GPA: 3.93/4.00 | Deans' Honour List**

Expected Graduation: April 2028

Coursework: Electronics and Instrumentation | Embedded Systems | Data Structures & Algorithms | Signals and Systems

EXPERIENCE

Embedded Systems Engineer (Co-op)

Cambridge, Ontario

Rockwell Automation – Allen-Bradley

Jan 2026 – Dec 2026

- Designed and programmed a **thermocouple DAQ board** for systems up to **600 V**, maintaining **<2°C accuracy** and **20 kHz current sampling** for RMS measurement.
- Designed the **third-revision power-cell control board** in Siemens Xpedition for Rockwell Automation **PF6000 medium-voltage drives**, integrating a co-processor and functional-safety hardware while reducing board size
- Designed and validated **PF6000 simulation interface hardware in KiCad**, including debugging and board bring-up
- Developed a Python/CIP automation tool for remote FPGA and fiber-optic parameter management, cutting motor-drive test setup time from **5 hours to ~10 minutes** across 4 global R&D teams

Sensing & Actuation Lead

Hamilton, Ontario

McMaster Exoskeleton

Oct 2024 – Current

- Led an **11-member sensing and actuation team** developing an **800 Hz real-time control pipeline** across **8 IMUs and 4 motors**, implementing multi-stage **Kalman and complementary filtering** for motion estimation and actuator feedback
- Designed a **4-layer, 48 V / 50 A power distribution board** in Altium, incorporating high-current power routing, thermal design, and controlled emergency discharge for system shutdown
- Led electrical debugging and failure isolation during competition integration for power-distribution and sensing issues

LV Electronics / Firmware Team Member

Hamilton, Ontario

Mac Formula Electric (FSAE)

Oct 2025 – Current

- Migrated **STM32 dashboard to LVGL 9.3**, cutting flash usage by **176 KB / 10.4%** through firmware optimizations
- Developed a **modular LVGL StatusBar** framework for persistent real-time fault status across all **embedded UI screens**
- Added **real-time CAN telemetry** displays for inverter states, temperatures, pack voltage, and discharge current, enabling **rapid fault diagnosis** and **live subsystem validation** during testing

Engineering Computing Teaching Assistant

Hamilton, Ontario

McMaster Engineering Faculty

Sep 2025 – Current

- Supported 30+ students in understanding file I/O, conditionals, loops, and **object-oriented programming**

PERSONAL PROJECTS

JamDeck - Winner, MakeUofT Hardware Hackathon

Toronto, Ontario

ESP32, C++, WiFi, JSON, MPU6050, FSR, Flex Sensor, LCD

Feb 2026

- Won **MakeUofT**, Canada's largest hardware hackathon, competing among **270 participants** with a real-time ESP32 music controller using **5 sensor/input systems**
- Developed **C++ firmware** to poll sensors, debounce buttons, update LCD UI, and transmit JSON events over WiFi
- Filtered noisy FSR/gyroscope** inputs using thresholding and event throttling to improve real-time responsiveness

Automated Wrestling Scoring System

Mississauga, Ontario

ESP32, Python, OpenCV, Bluetooth, Embedded C++

Jun 2025 – Jul 2025

- Designed an embedded scoring interface with **sub 100 ms** Bluetooth response using **ESP32**
- Built a modular preprocessing and classification pipeline achieving **~75% preliminary accuracy** on move detection

SKILLS

Hardware & Validation: PCB & Schematic Design, Board Bring-up & Debugging, Hardware Validation, Oscilloscopes, Soldering

Embedded Hardware and Protocols: ARM Cortex-M, STM32, ESP32, UART, SPI, I2C, CAN, EtherNet/IP, CIP, JTAG

Embedded Systems: FreeRTOS, Mbed OS, LVGL, MCU Architecture

Software & Design Tools: Altium Designer, Siemens Xpedition (Mentor Graphics), KiCad, LTspice, C/C++, Python, Simulink, Git

AWARDS & ACHIEVEMENTS

MakeUofT Winner | Hack the North Semifinalist | Deans' List (All Academic Terms)